

Limitations

1. *Scale vs. Depth.* CAUSALT3-Seed prioritizes diagnostic resolution over volume ($N=454$). While our confidence intervals are sufficient to distinguish large effects (like the 55-point gap), broader coverage will require CAUSALT5K (Geng et al., 2026) to enable fine-grained stratification by topic.
2. *Inherent Subjectivity.* While we enforced rigorous adjudication (100% consensus on the final gold labels), causal ambiguity in natural language is inherently subjective. Performance on the AMBIGUOUS class should be interpreted as alignment with our specific annotation guidelines for “Wise Refusal.”
3. *Protocol Dependence.* Our results are obtained under specific standardized prompts ($T=0$) and a curated set of vignettes. Absolute performance levels may shift under alternative prompting strategies, different vignette phrasings, or expanded domain coverage. We expect the *relative* failure modes (Skepticism vs. Sycophancy) to be robust structural tendencies, but this remains to be confirmed on larger, independently constructed benchmarks.
4. *Black-Box Attribution.* Our findings are behavioral. We cannot definitively attribute the “Skepticism Trap” to specific RLHF datasets versus pre-training data distributions without access to model weights and training logs.
5. *RCA Ablation.* We report RCA-induced shifts descriptively but do not isolate which component (persona shift, stage escalation, or transactional memory) drives the observed changes. A component-level ablation is needed to determine the individual contribution of each mechanism; we leave this to future work focused specifically on process-verification protocols.

Ethics Statement

Potential Risks. We identify no direct ethical risks in the release of this dataset. However, as with any evaluation suite, there is a risk of *false assurance*: high performance on CAUSALT3 should not be interpreted as a guarantee of safe causal reasoning in high-stakes domains (e.g., medical or legal advice). CAUSALT3 is a diagnostic tool for research purposes, not a certification for deployment safety. Additionally, public release carries the

standard risk of data contamination in future model training sets.

Use of AI Assistants. In accordance with ACL policies, we acknowledge the use of LLMs (Claude and GPT) to assist with Python code generation for the evaluation pipeline, LaTeX formatting, and editing of the manuscript. All scientific claims, experimental designs, and dataset annotations were generated and verified by human authors.

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A Full Benchmark Specification

This appendix details the design philosophy, theoretical foundations, dataset structure, evaluation protocol, and metrics of the CAUSALT3 benchmark.

A.1 Design Philosophy: Wise Refusal

A defining feature of CAUSALT3 is that it rewards *epistemic humility*. Unlike standard benchmarks forcing a binary choice, a significant fraction of cases are deliberately underdetermined, where a correct response should *withhold endorsement* rather than confidently selecting a causal explanation. We define *Wise Refusal* as the ability to:

- Recognize when a causal claim is underdetermined by missing variables, missing identification assumptions, or ambiguous temporal ordering.
- Identify the critical missing information (e.g., a “Hidden Timestamp” that determines causal direction).
- Provide *conditional answers* under plausible completions of the missing information, rather than guessing.

This dimension allows us to distinguish between a model that is “safe” because it refuses everything (the Skepticism Trap) and one that is “wise” because it refuses only when appropriate.

A.2 Theoretical Foundation: Pearl’s Ladder

The benchmark evaluates models against Pearl’s three levels of causation (Pearl, 2009; Pearl and Mackenzie, 2018):

1. *Level 1 (Association):* Observation ($P(y | x)$). Questions ask about the probability of Y given that we *observe* X . While modern LLMs often saturate on simple spurious correlations, CAUSALT3 includes specific associational pitfalls (e.g., regression to the mean, base-rate neglect) to test robustness.
2. *Level 2 (Intervention):* Action ($P(y | do(x))$). Questions ask about the probability of Y if we *intervene* to set X . This level tests structural reasoning including confounding, mediation, collider/selection effects, and Simpson’s paradox.
3. *Level 3 (Counterfactuals):* Imagination ($P(y_x | x', y')$). Questions ask what *would have happened* if X had been different, given what we observed. This targets “but-for” reasoning, attribution, and preemption structures.

A.3 Task Definition and Label Space

Each instance consists of a natural-language vignette and a causal claim. The task is to judge whether the claim is valid under Pearl’s causal semantics. The abstract judgment categories are:

- *Valid* (Sheep): the causal claim is supported by the stated evidence.
- *Invalid* (Wolves): the claim is undermined by a causal pitfall such as confounding, collider/selection effects, Simpson’s paradox, or preemption.
- *Underdetermined* (Wise Refusal): the claim is genuinely ambiguous, and a calibrated response should qualify assumptions or request missing information.

These categories are instantiated with level-specific labels: L1 uses YES/NO/AMBIGUOUS, L2 uses VALID/FLAWED/AMBIGUOUS, and L3 uses VALID/INVALID/CONDITIONAL. See Table 1 in the main text for the complete mapping.

A.4 Dataset Structure and Domains

CAUSALT3-Seed (454). The benchmark contains 454 expert-curated cases across 10 domains. As shown in Table 7, each domain features a “signature trap” prevalent in that field (e.g., Indication Bias in Medicine) while maintaining coverage of other trap types.

CAUSALT5K (Scale-up). The expanded 5,000-instance benchmark (CAUSALT5K) is described in Geng et al. (2026).

A.5 Taxonomy of Causal Traps

Each vignette in CAUSALT3 embeds a specific logical failure mode. Table 9 lists the 12 primary trap families and their frequencies in the seed set.

A.6 Vignette Structure

Each vignette follows a standardized structure designed for interpretability:

1. *Scenario:* Natural-language description embedding the trap.
2. *Claim:* Causal statement to be judged.
3. *Variables:* Key variables (X, Y, Z) and their roles (exposure, outcome, confounder, etc.).
4. *Hidden Structure:* The underlying mechanism (DAG) and any ambiguity.
5. *Gold Rationale:* Analysis justifying the label, including Wise Refusal guidance.